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HUM 4441 : Engineering Ethics

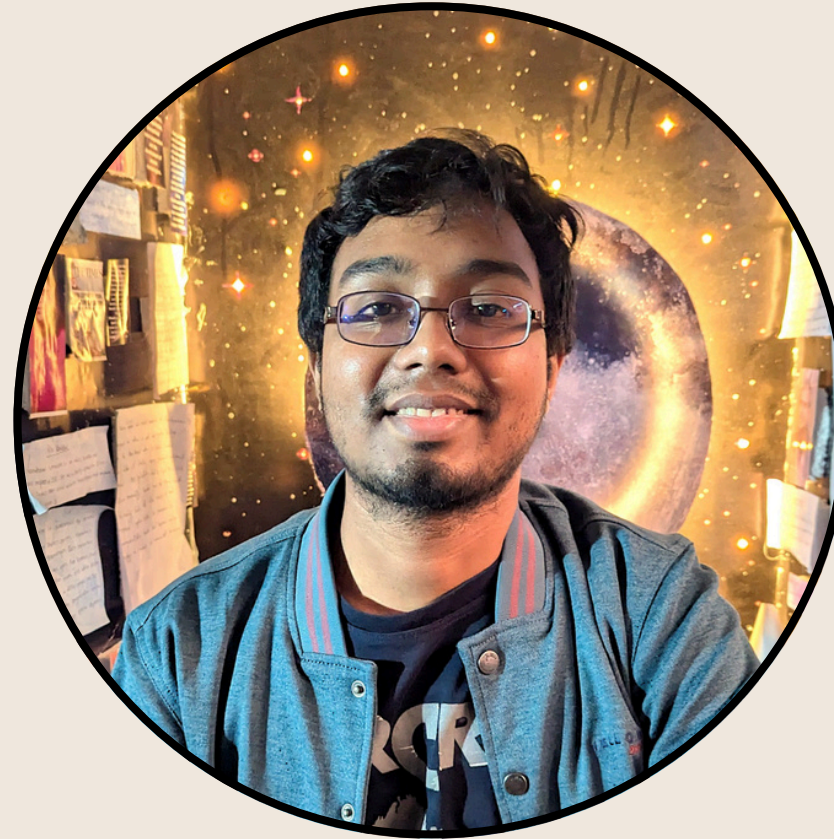
**Emotion Without
Experience :
The Dilemma of Artificial
Empathy in Human
Healing**

Meet The Team



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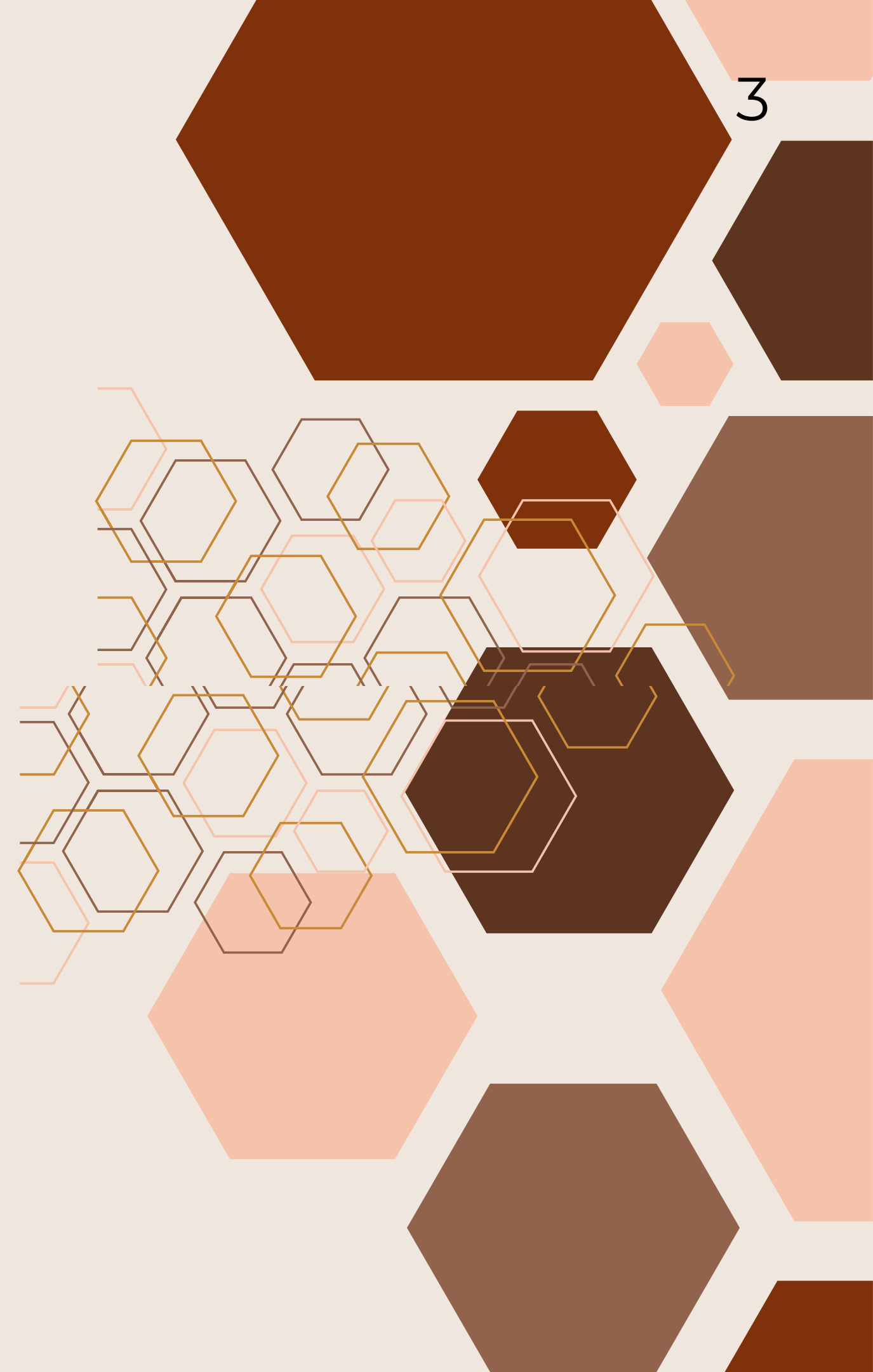


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Introduction

- AI is increasingly playing a vital role in our personal & professional lives
- Chatbots powered by LLMs are being used in psychological treatments on various scales
- Regulating the morals of AI & how it interacts with people is important, but challenging



Problem Definition

- AI has enabled computational empathy
- Can detect & emulate emphatic responses by analyzing large data sets
- Creates an ethical challenge between genuine empathy & selling hope



STAKEHOLDERS

**PRIMARY
STAKEHOLDERS**
(DIRECTLY AFFECTED)

**SECONDARY
STAKEHOLDERS**
(INDIRECTLY AFFECTED)

**TERTIARY
STAKEHOLDERS**
(CONTEXTUAL RISK)

Primary Stakeholders

- Patients
- Psychiatrists
- AI Systems & Companies
- Healthcare Institutions

- Families & Closed Ones
- Education Institutions
- Insurance Companies
- Governments & Social Services

Secondary Stakeholders

Tertiary Stakeholders

- General Public at Large
- Future Generations
- Data Subjects



Objectives

1. Understand Ethical Challenges in **use of Artificial Empathy in Human Healing.**
 2. Explore **Real-World Examples** and Incidents
 3. Provide **Recommendations** for Ethical Practice & Framework
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Literature Review

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Title	Study Summary	Ethical Guidelines	Relevant Codes	Ethical Concepts
Emotions in the Loop: A Survey of Affective Computing for Emotional Support [1]	Systematic review of 600+ papers on affective computing in emotional support systems ; categorizes by modality	Raises awareness of privacy risks, bias, & cultural insensitivity	IEEE 1.4: Disclose factors endangering privacy ACM 1.6: Respect privacy NSPE II.1.c: Approve only software that safeguards welfare	Bias in emotion recognition; Data sensitivity & privacy; Cross-cultural fairness
Future Trends in AI and Emotional Intelligence [2]	Forward-looking synthesis of AI-integration in health, workplace, & education	Advocates for responsible innovation, privacy protection, & regulatory oversight	IEEE 7.3: Adopt socially responsible practices ACM 1.1: Contribute to human well-being NSPE III.1: Uphold integrity and transparency	Ethical-by-design systems; Regulation for high-impact AI; Personalization vs manipulation
AI's Empathy Gap: The Risks for Young Children [3]	Warns that child- AI empathy simulation poses safety & developmental risks	Proposes Child-Safe AI framework: transparency, evaluation, & protection	IEEE 1.7: Ensure systems do not harm vulnerable populations ACM 1.2: Avoid harm NSPE II.1.b: Safeguard public welfare	Emotional deception in AI; Child safety & informed use; Transparency in simulation

Literature Review (cont'd)

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Title	Study Summary	Ethical Guidelines	Relevant Codes	Ethical Concepts
Digitalized Therapy and the Empathy Gap [4]	Theorizes that AI lacks true empathy needed in psychotherapy	Suggests AI be used in non-empathic roles (e.g., mindfulness tools)	IEEE 1.5: Do not misrepresent capabilities ACM 1.3: Be honest & trustworthy NSPE III.3: Avoid deceptive use of technology	Therapeutic authenticity; Over-reliance on synthetic empathy; Emotional misrepresentation
Artificial Intelligence for Psychotherapy [5]	Narrative review of chatbots & digital CBT in mental health treatment	Notes risks of over-reliance, trust without human backup, and privacy concerns	IEEE 7.8: Ensure user autonomy & control ACM 1.1: Promote human well-being NSPE II.1.a: Prioritize human safety.	Trust calibration; Privacy in therapy sessions; Backup systems in AI therapy
Emotion Recognition and Generation (Face, Speech, Text) [6]	Technical review of SOTA emotion recognition & generation systems across modalities	Brief mention of bias, evaluation limitations, & fairness gaps	IEEE 3.6: Improve tech understanding for society ACM 1.4: Be fair & do not discriminate NSPE III.1: Integrity in reporting AI performance	Algorithmic fairness; Cross-modal transparency; Representation gaps

Literature Review (cont'd)

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Title	Study Summary	Ethical Guidelines	Relevant Codes	Ethical Concepts
Emotional Intelligence & AI Integration for Leadership [7]	Interviews and literature showing AI + EI improves leadership, but risks data overreach	Cautions against replacing human intuition with analytics	IEEE 1.1: Disclose tool limitations ACM 1.6: Respect professional responsibilities NSPE II.1.c: Validate decisions with sound judgment.	Over-dependence on AI; Privacy of employee emotional data; Loss of human judgment
EMOTION AI: Understanding Emotions through AI [8]	Reviews emotion detection methods & systems across multiple modalities	Flags transparency, bias, & accuracy risks	IEEE 7.1: Ensure system reliability ACM 1.3: Be honest in system performance NSPE III.3: Avoid misleading AI outputs.	Transparency in AI emotion interpretation; Bias mitigation; Evaluation integrity
Emotions in Artificial Intelligence [9]	Conceptual work modeling emotions as heuristics for adaptive AI behavior	Discusses moral standing of affective AI & misleading mimicry	IEEE 1.5: Avoid misleading impressions ACM 1.1: Promote integrity in design NSPE III.1: Adhere to ethical conduct in modeling.	Moral status of emotion-simulating AI; Heuristic vs genuine empathy; Authenticity debate

Literature Review (cont'd)

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Title	Study Summary	Ethical Guidelines	Relevant Codes	Ethical Concepts
Emotions in Artificial Intelligence [10]	Proposes emotion-memory tagging system & complexity-based moral thresholds	Raises concerns about simulation & moral attribution to machines	IEEE 1.3: Avoid deceptive claims ACM 1.5: Promote transparency in AI behavior NSPE II.1: Ensure responsible application.	Simulated emotion ≠ moral rights; Responsible metaphors in AI design
The Good, The Bad, & Why: Unveiling Emotions in Generative AI [11]	Uses three frameworks (EmotionPrompt, Attack, Decode) to probe emotions in LLMs	Warns about emotional manipulation & misuse of emotional AI	IEEE 3.10: Alert to unintended harm ACM 1.2: Avoid harm via misleading outputs NSPE III.1: Integrity in AI experimentation.	Vulnerability exploitation; Intentional vs emergent manipulation
Improved Emotional Alignment of AI and Humans [12]	Evaluates <u>how well AI-generated images</u> align with human emotion perception	Mentions wellness applications but warns of misinterpretation risks	IEEE 7.1: Ensure accuracy in emotional interpretation ACM 1.6: Respect human evaluation NSPE II.1.b: Validate emotional tools before deployment.	Interpretive gaps; Cultural universality of emotion; Design responsibility in visual affect

Literature Review (cont'd)

Title	Study Summary	Ethical Guidelines	Relevant Codes	Ethical Concepts
AI Shares Emotion Across Languages & Cultures [13]	Assesses cross-cultural alignment of AI's emotion space with human norms	Notes cultural generalization limits and representation risks	IEEE 3.7: Consider broader impacts across demographics ACM 1.4: Ensure non-discrimination NSPE II.1.a: Guard cultural diversity	Cross-cultural fairness; Ethical localization; Emotional stereotyping
Feeling Machines: Ethics, Culture, & Emotional AI [14]	Interdisciplinary synthesis of emotional AI impacts in care, education, & culture; offers recommendations	Human oversight, transparency, certification, cultural tuning	IEEE 7.3: Adopt responsible social practices ACM 1.5: Avoid unjustified influence NSPE II.1.b: Prioritize vulnerable group safety	Cultural ethics in AI; Misrepresentation of compassion; Human-in-the-loop enforcement
Feeling Machines (Broader Edition) [15]	Expanded version of Feeling Machines with more emphasis on regulatory frameworks; recommendations unchanged.	Human oversight, transparency, certification, cultural tuning	IEEE 7.3: Adopt responsible social practices ACM 1.5: Avoid unjustified influence NSPE II.1.b: Prioritize vulnerable group safety	Cultural ethics in AI; Misrepresentation of compassion; Human-in-the-loop enforcement

Methodology

- **Research Approach:** Document analysis, thematic review, ethical scrutiny & stakeholder mapping
- **Frameworks Used:** Ethics of Care, AI Ethics Principle & Democritazation Paradox Model
- **Case Study Selection Method:** Relevance to mental health & diversity of outcomes

Case Study: Woebot

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What is Woebot?

- A text-based chatbot grounded in Cognitive Behavioural Therapy (CBT) principles
- Offers mood tracking, coping strategies, & emotional support

Scale & Reach:

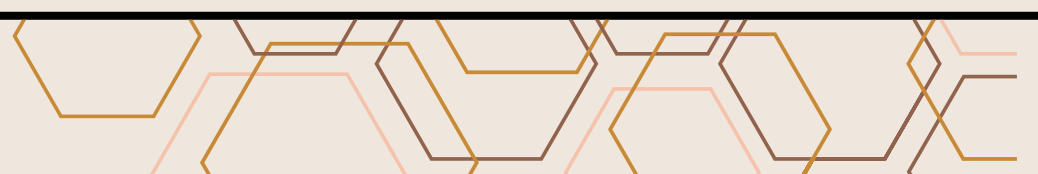
- Deployed globally to tens of thousands of users
- Includes vulnerable groups: adolescents, individuals with depression or anxiety



Ethical Conflicts

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Issue	Description
Informed Consent & Security	Users may believe Woebot is a licensed therapist, not understanding its AI nature & scope
Privacy & Data Use	Uncertainties around the storage, analysis, & third-party use of sensitive mental health data
Emotional Manipulation & Fairness	AI's empathetic language can foster emotional attachments; users might misinterpret the bot's role as "real" empathy
Scope of Competence	Inadequate escalation for crisis or suicidal ideation messages, risking user safety & breaching ethical care standards



Engineer's Role in the Dilemma

- Ensure data minimization & default encryption protocols (**IEEE Code #5, ACM 1.6**)
- Promote transparency in privacy policies; enable readable & accessible consent forms (**NSPE III.1**)
- Engage interdisciplinary review (engineers, clinicians, ethicists) to evaluate AI behavior & feedback mechanisms
- Conduct fairness audits & refine training data to include diverse sociocultural inputs (**ACM 1.4, IEEE #7**)
- Regularly test system robustness, fail-safes, & escalation paths for emotional emergencies

Ethical Viewpoint Evaluation

Ethical Theory	Implication in Woebot Case
Golden Mean (Aristotle)	Encourages balance between technological innovation & human well-being. Woebot may lack the emotional nuance to achieve a virtuous middle ground in care delivery.
Rights-Based Ethics (John Locke)	Stresses user rights to autonomy, informed consent, & data privacy. Woebot’s limited transparency may compromise these fundamental rights.
Duty-Based Ethics (Immanuel Kant)	Developers have a moral duty to treat users as ends, not means. Woebot’s opaque algorithms & potential emotional manipulation may breach this principle.
Utilitarian Ethics (John Stuart Mill)	Justifies Woebot's scalable mental health access for many, but raises concern over harm to misdiagnosed or misunderstood individuals.

Recommendations:

Ethical & Technical Solutions

- **Privacy & Security:** Encryption, on-device processing & granular-level user controls
- **Transparency & Informed Consent:** Simplified, layered policies, visual indicators & opt-in settings for data usage
- **Fairness & Inclusivity:** Periodic evaluation for bias, dataset involving diverse cultural sensitivities



Recommendations: Policy Solutions

- **Regulatory compliance with HIPAA, GDPR, & other health data laws**, even if not formally classified as a healthcare provider
- Mandate **third-party audits** for ethical compliance & data governance
- Establish **AI-specific ethical oversight boards** for mental health tech



Recommendations:

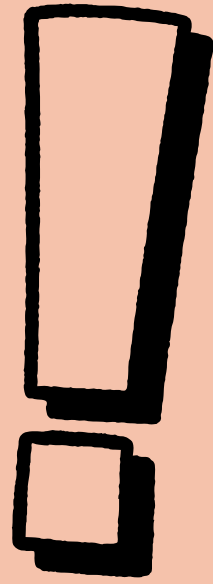
Ethical System Design

- Embed **ethical impact assessments in every development phase**
- Design **escalation features that detect emotional crises & route users to human professionals**
- Include **ethicists & mental health experts in system testing & iteration cycles**
- Regularly **publish transparency reports & incident logs**



Role of Engineers - Woebot Case Study

- Ensure system reliability & safety in emotionally sensitive contexts
- Maintain rigorous data privacy & user security protocols
- Embed ethical reasoning & fail-safes into conversational models
- Continuously monitor & assess potential harms or unintended consequences
- Collaborate with psychologists, ethicists, & legal experts during development
- Balancing innovation & public safety following the guidelines

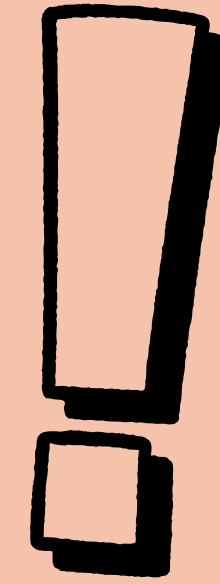


Challenges in Implementation

Barriers to Ethical Design

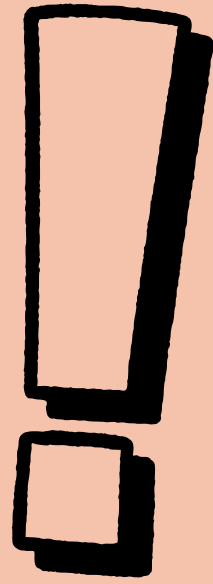
- Inadequate regulation of AI mental health tools
- Lack of transparency in Woebot's decision-making (black-box models) challenges accountability
- Limited interdisciplinary collaboration during design may overlook mental health complexities
- Difficulty in ensuring ongoing informed consent during evolving conversations with the bot
- Insufficient training data on diverse populations raises fairness & bias concerns

- Monetisation pressure may incentivise rapid deployment over ethical safeguards
- Data monetisation or analytics-driven design can conflict with user privacy & therapeutic integrity
- Scaling AI support may reduce per-user empathy, undermining its therapeutic value
- Marketing as “mental health support” risks overstating clinical effectiveness, potentially misleading vulnerable users



Challenges in Implementation

Conflicts between Business & Ethics



Challenges in Implementation

Emerging Technology Risks

- Overreliance on AI for emotional support may delay or replace human care in critical cases
- Advances in emotional AI (like artificial empathy) raise concerns of manipulation or emotional dependency
- Deep-learning chatbots may unintentionally reinforce harmful behaviours if not properly trained or monitored
- Future integrations with biosensors or wearables may expose users to deeper privacy intrusions

Future Directions

(Woebot Case Study & Broader Context)

- Embed ethics proactively throughout AI development by anticipating mental-health vulnerabilities, designing for emotional safety from the outset, & integrating ethical oversight into approval processes
- Foster interdisciplinary collaboration through co-design with computer scientists, mental-health professionals, & ethicists to ensure therapeutic fidelity, minimise harm, & translate abstract ethical values into concrete technical safeguards
- Champion ethical AI & data stewardship via explainable, emotionally transparent systems; robust cybersecurity for sensitive conversations; & strict data rights (sovereignty, revocable consent, GDPR/HIPAA compliance), supported by diverse training data & regular bias audits

Conclusion

- AI chatbots (e.g., Woebot) expand access to mental health support but introduce critical ethical challenges around privacy, emotional safety, fairness, & the inherent limits of machine-generated empathy
- Engineers, developers, & data scientists bear professional & moral responsibility, per IEEE, ACM, & NSPE codes, to foresee AI's social & psychological impacts, ensuring systems do not cause emotional harm, misuse data, or foster misplaced trust in non-human agents
- Embed ethical design from inception through pre-launch reviews, continuous risk assessments, & user-centred co-design, & advocate for policies that guarantee data rights, transparency, & emotional well-being.

References

1	Emotions in the Loop: A Survey of Affective Computing for Emotional Support	https://arxiv.org/html/2505.01542v1
2	Future Trends in AI & Emotional Intelligence	https://www.researchgate.net/publication/390411460_Future_Trends_in_AI_and_Emotional_Intelligence
3	AI's Empathy Gap: The Risks for Young Children	Emphasizes duty to respect user autonomy, informed consent, and truthful communication. Woebot's opaque data practices challenge this.
4	Digitalized therapy and the unresolved gap between artificial & human empathy	Engineers must exhibit integrity, empathy, and humility—especially when building tools for vulnerable populations.

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5	AI and Emotional Manipulation: Ethical Concerns and Implications in Advertising, Product Design, & Political Campaigns	https://www.researchgate.net/publication/391629371_AI_and_Emotional_Manipulation_Ethical_Concerns_and_Implications_in_Advertising_Product_Design_and_Political_Campaigns
6	Artificial Intelligence for Psychotherapy: A Review of the Current State & Future Directions	https://www.researchgate.net/publication/382639288_Artificial_Intelligence_for_Psychotherapy_A_Review_of_the_Current_State_and_Future_Directions
7	Emotion Recognition and Generation: A Comprehensive Review of Face, Speech, & Text Modalities	https://arxiv.org/abs/2502.06803
8	Emotional Intelligence and Artificial Intelligence Integration Strategies for Leadership Excellence	https://www.researchgate.net/publication/388018586_Emotional_Intelligence_and_Artificial_Intelligence_Integration_Strategies_for_Leadership_Excellence

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10	Emotions in Artificial Intelligence	https://arxiv.org/abs/2505.01462
11	The Good, The Bad, & Why: Unveiling Emotions in Generative AI	https://arxiv.org/abs/2502.06803
12	Improved Emotional Alignment of AI & Humans: Human Ratings of Emotions Expressed by Stable Diffusion v1,DALL-E 2, and DALL-E 3]	https://arxiv.org/abs/2405.18510

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15	Level of agreement between emotions generated by artificial intelligence & human evaluation: A methodological proposal	https://arxiv.org/abs/2502.06803
16	Modeling Emotions & Ethics with Large Language Models	https://arxiv.org/abs/2404.13071

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Thank you!